DAY 1 | VIRTUAL MACHINE HANDS-ON ACTIVITY

Network configuration, service enumeration, and security monitoring

Laboratory environment: Oracle VirtualBox, Kali Linux, Metasploitable 2, SIEMv2 (Suricata, Kibana, Arkime).

Objective: I configured the virtual network, verified communication between laboratory hosts, enumerated services on the designated vulnerable VM, and examined the corresponding security telemetry and alerts.

Evidence: I documented my setup and results below. Evidence Step 2 is intentionally excluded.

Kali internal address: 10.10.1.10/24

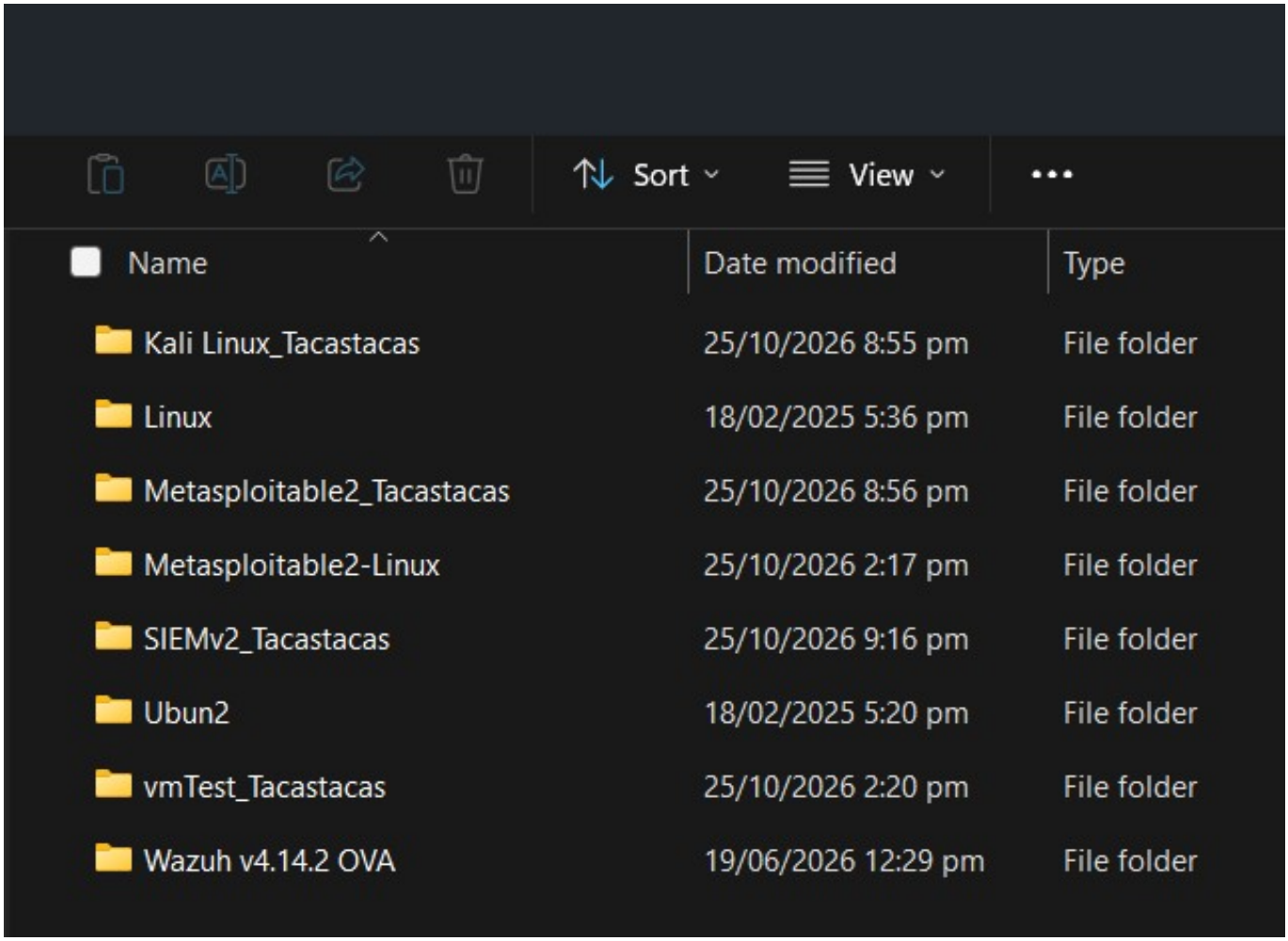
Metasploitable address: 10.10.1.20/24

SIEM NAT address: 10.0.2.4/24 during verification

Scan alert: ET SCAN Possible Nmap User-Agent Observed (SID 2024364)

STEP 1 | VM files and workspace

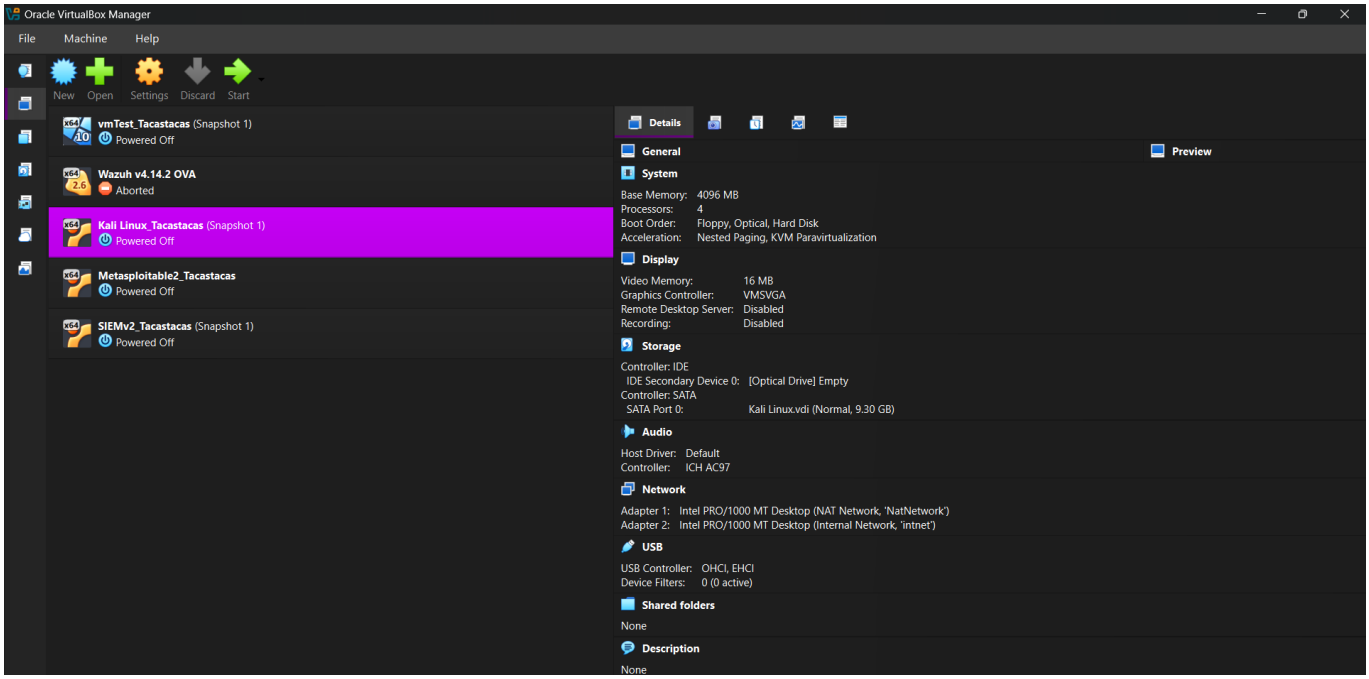
I organized the virtual machine files within my VirtualBox VM directory. The File Explorer evidence shows the VM folders on the C: drive rather than the D: drive specified in the original activity.



VirtualBox VM folders (personal account path obscured).

STEP 3 | VirtualBox Manager

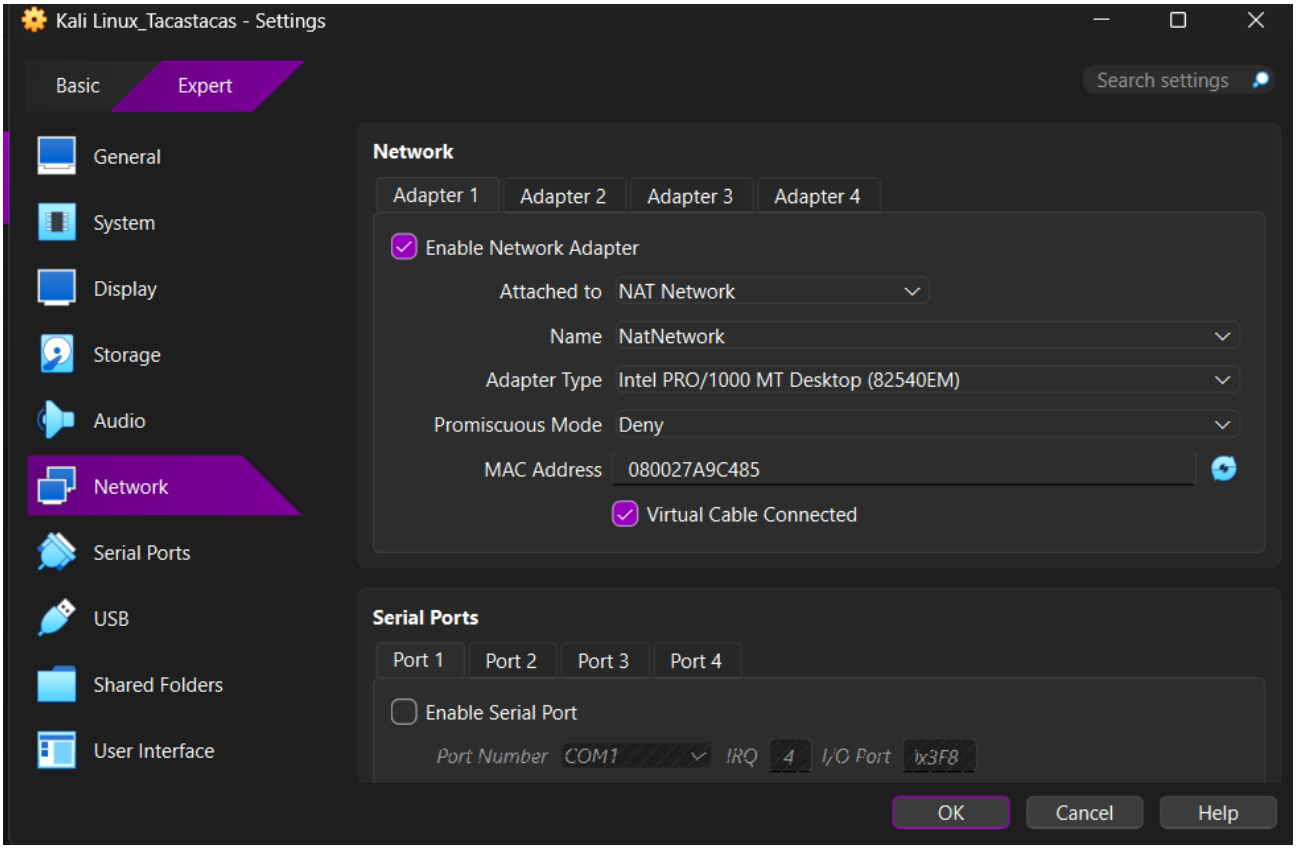
I created and configured the Kali, Metasploitable 2, and SIEMv2 virtual machines in Oracle VirtualBox.



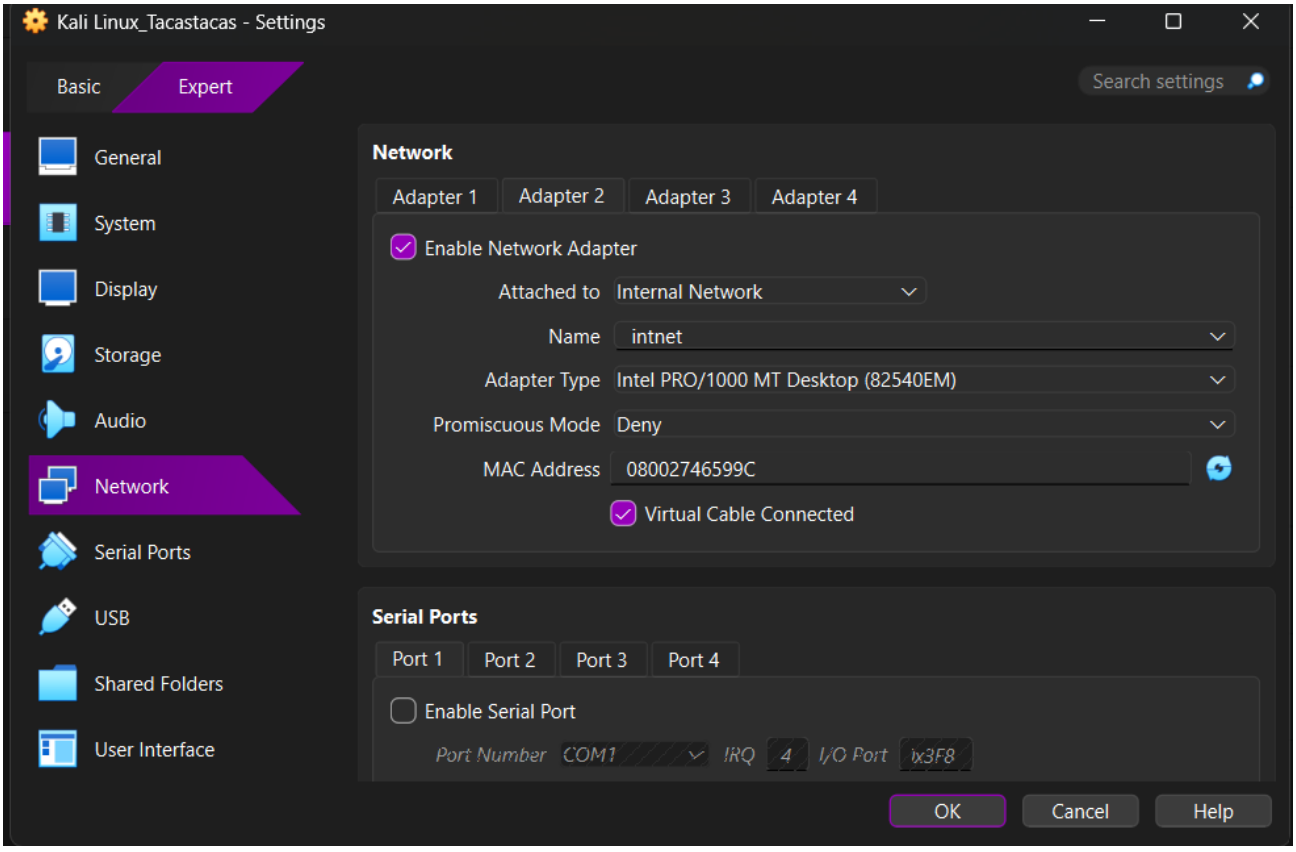
VirtualBox Manager showing the Kali Linux VM and the other laboratory machines.

STEP 4 | Kali Linux network adapters

I configured Adapter 1 to use NatNetwork and Adapter 2 to use the isolated internal network named intnet.



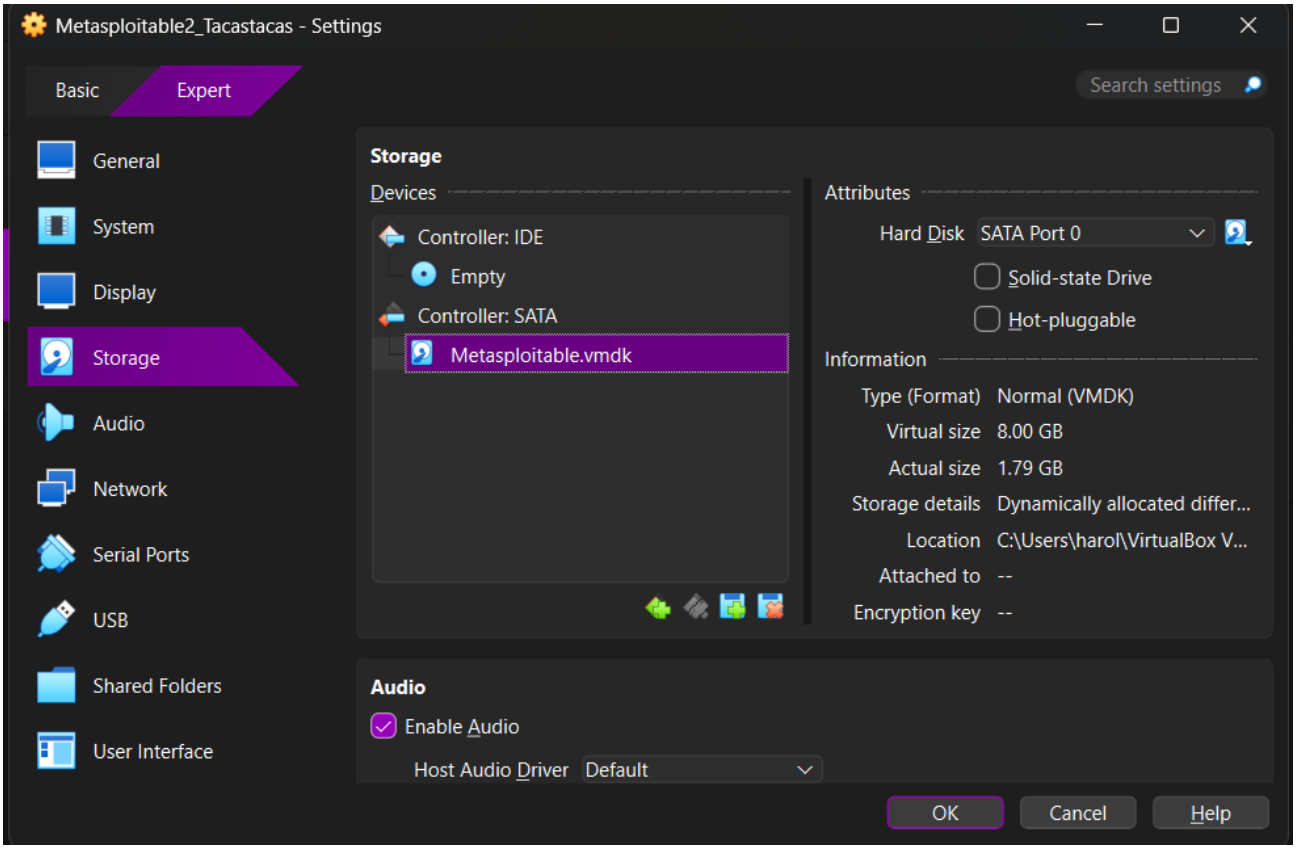
Adapter 1: NAT Network, NatNetwork.



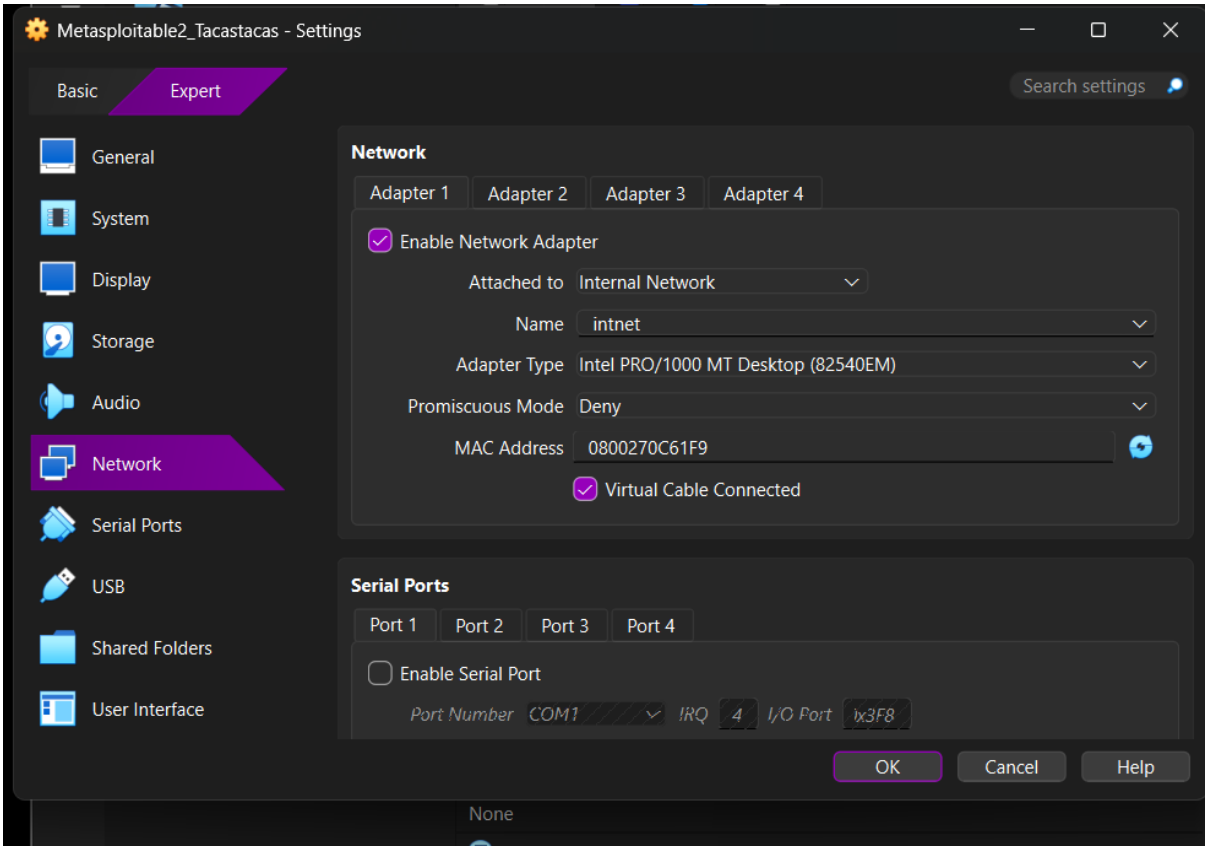
Adapter 2: Internal Network, intnet.

STEP 5 | Metasploitable virtual disk and network

I attached the Metasploitable.vmdk disk image and configured Adapter 1 to use the intnet internal network.



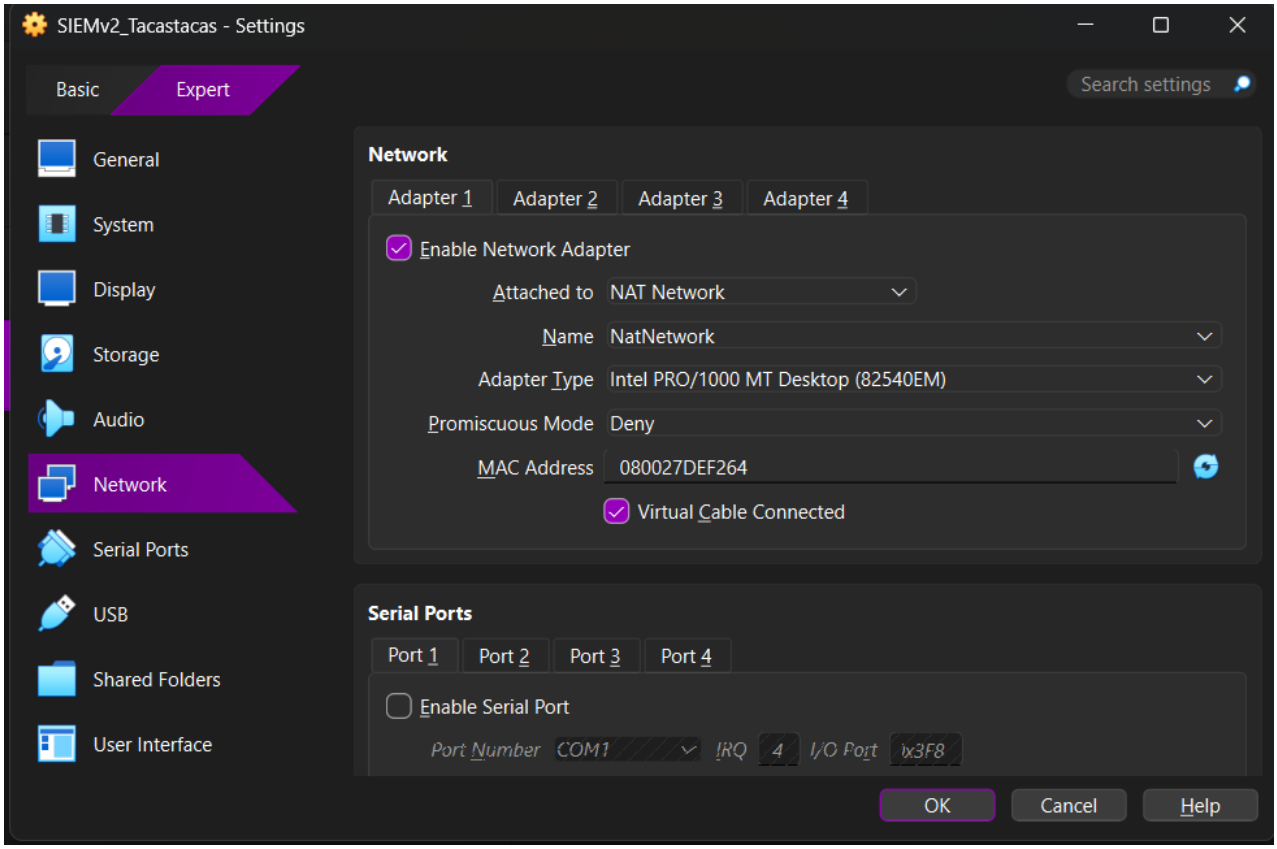
Metasploitable 2 storage: Metasploitable.vmdk attached.



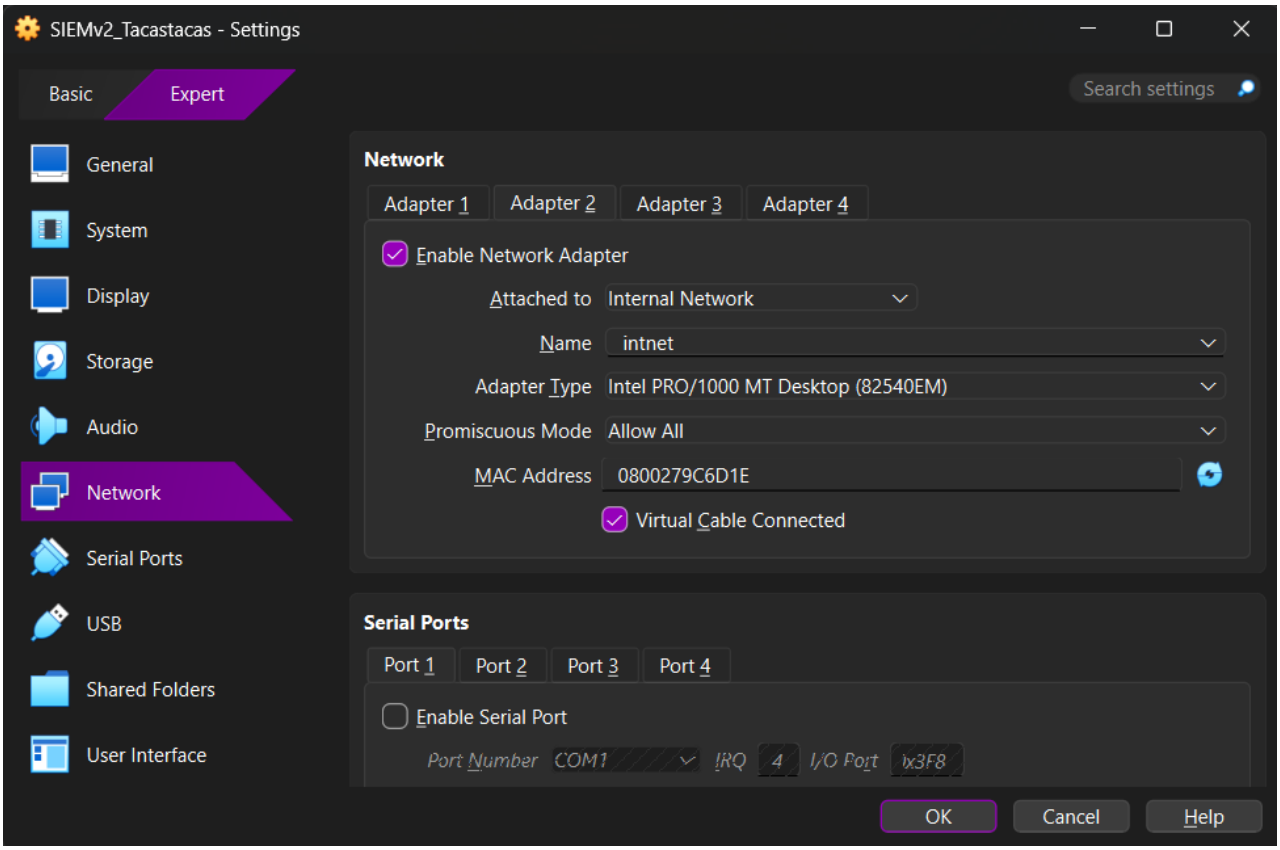
Metasploitable 2 network: Adapter 1, Internal Network intnet.

STEP 6 | SIEMv2 network configuration

I configured the SIEM with a NAT Network adapter and a second adapter on intnet, with promiscuous mode set to Allow All for network monitoring.



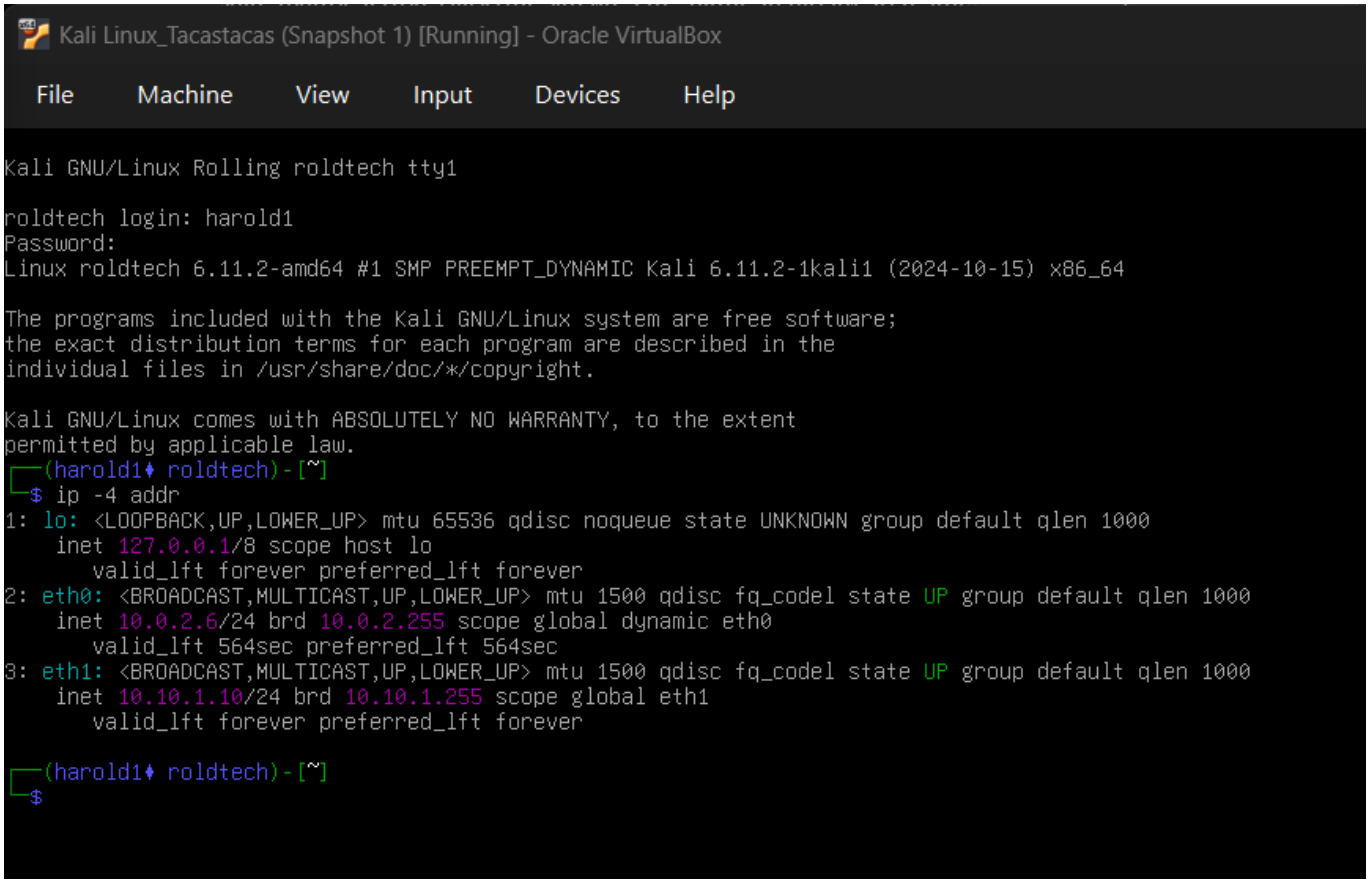
SIEMv2 Adapter 1: NAT Network, NatNetwork.



SIEMv2 Adapter 2: Internal Network intnet, Promiscuous Mode: Allow All.

STEP 7 | Kali static interface address

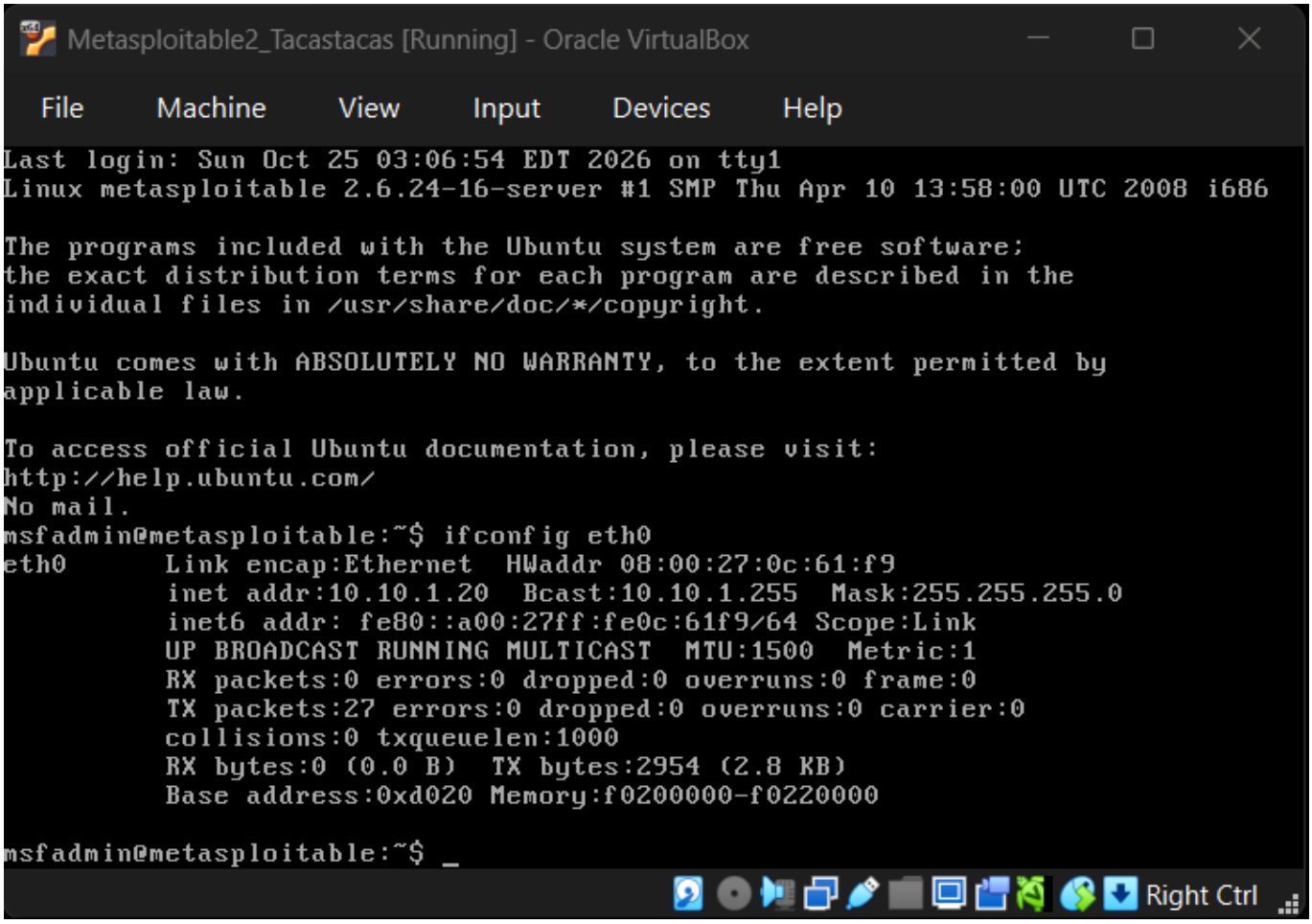
I checked the second Kali interface using ip -4 addr. The eth1 interface had the static address 10.10.1.10/24.



Kali interface verification: eth1 = 10.10.1.10/24.

STEP 8 | Metasploitable IP address

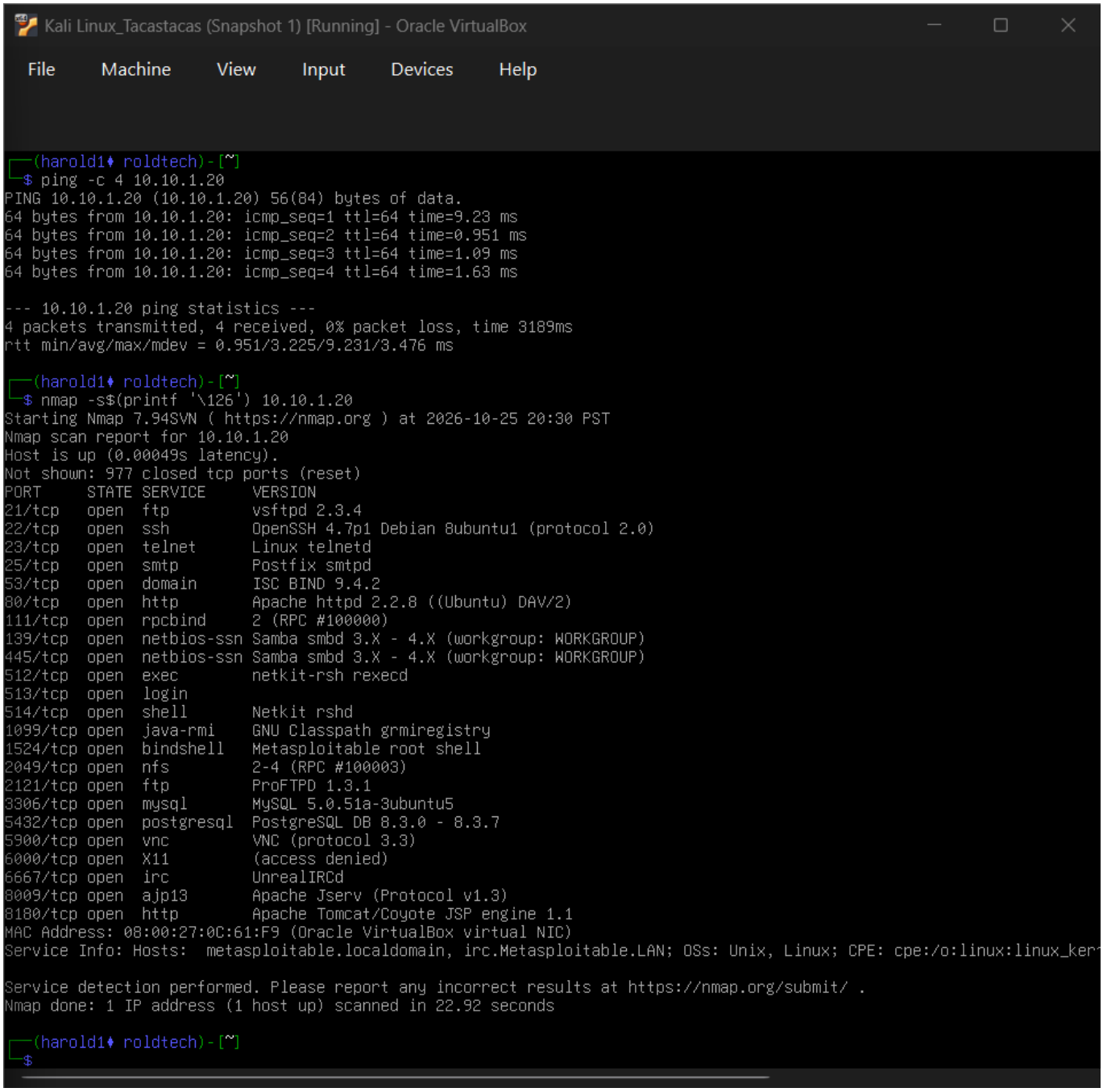
I checked the target interface with ifconfig eth0 and verified that it was configured as 10.10.1.20/24.



Metasploitable eth0 = 10.10.1.20.

STEP 9 | Kali-to-Metasploitable connectivity

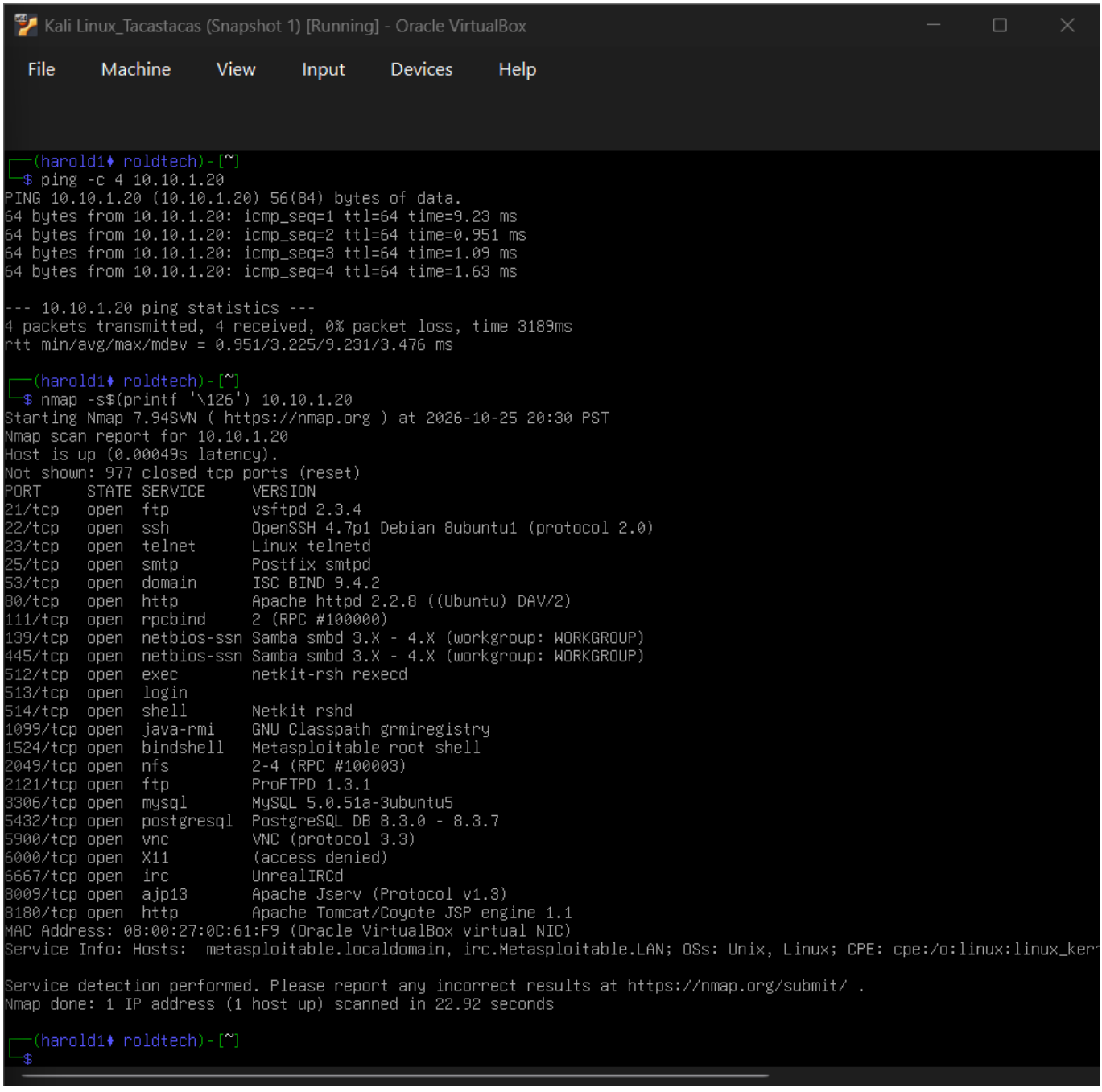
I ran ping -c 4 10.10.1.20 from Kali. All four ICMP echo requests received replies, with 0% packet loss.



Successful ping appears above the Nmap results in the terminal screenshot.

STEP 10 | Nmap service-version scan

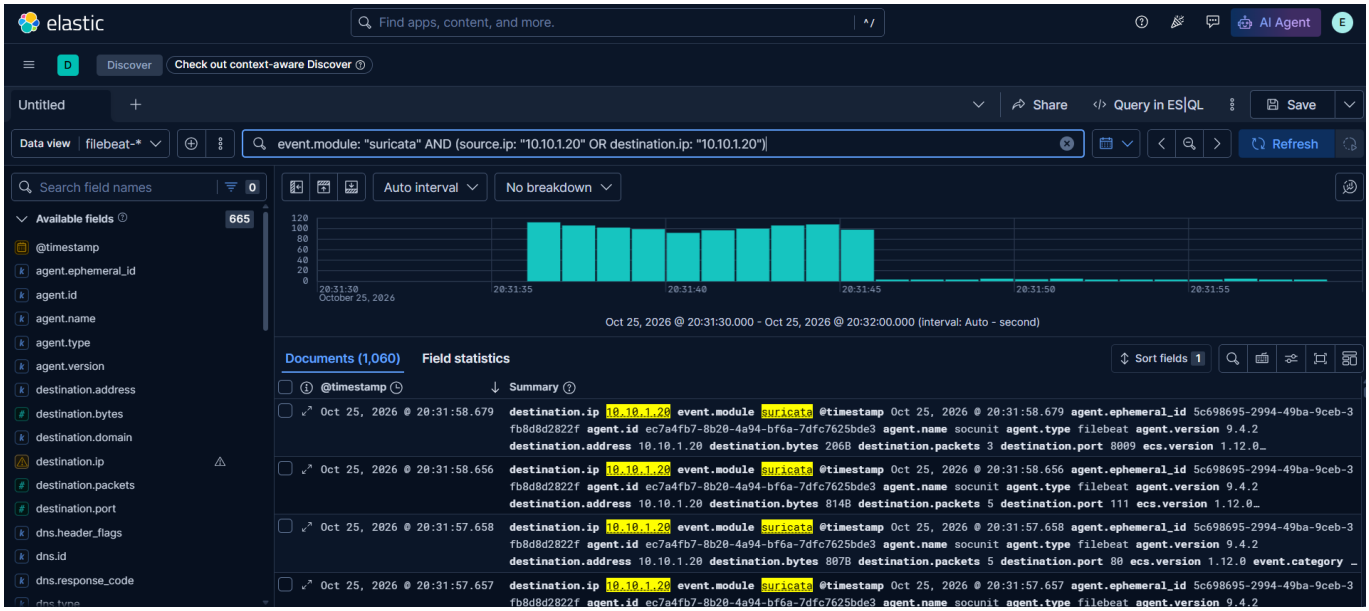
I performed Nmap service-version detection against 10.10.1.20. The scan showed 23 open TCP ports, including FTP (21), SSH (22), HTTP (80), and MySQL (3306).



Nmap -sV equivalent command, open ports, service versions, and scan completion.

STEP 11A | Kibana access and captured events

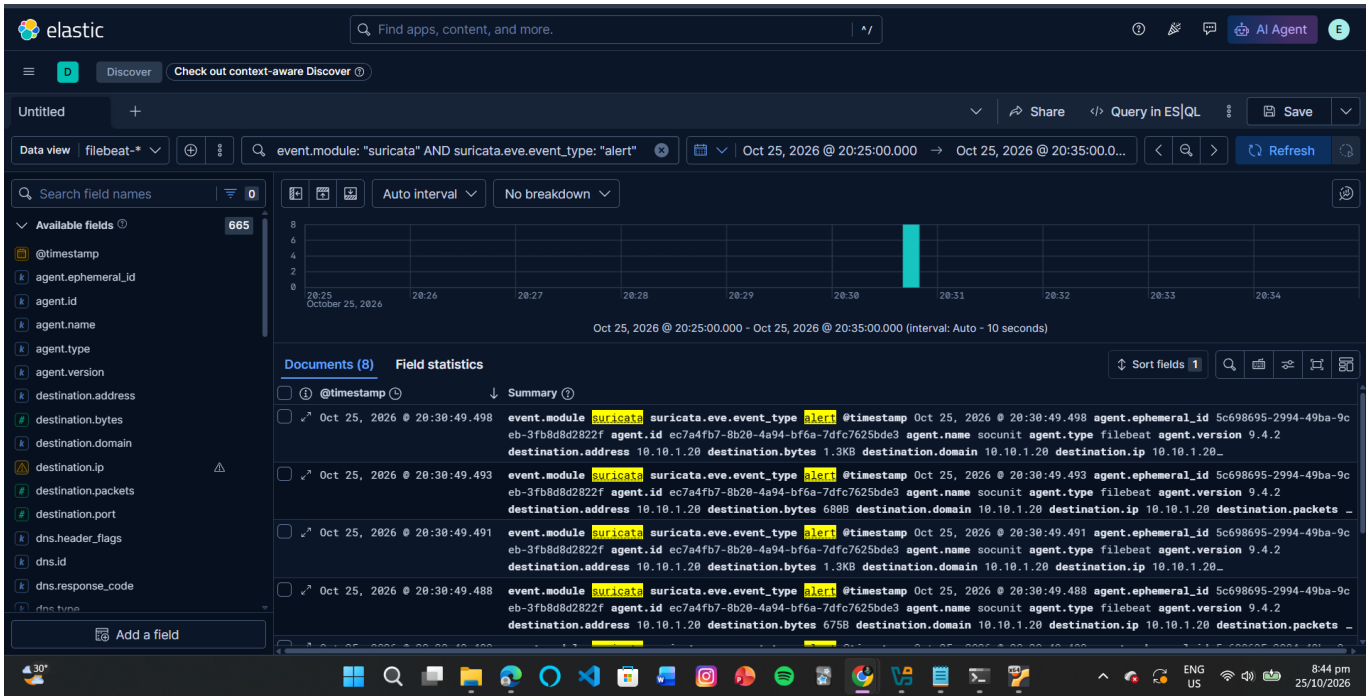
I opened Kibana Discover with the filebeat-* data view and verified that Suricata ingested traffic from 10.10.1.10 to 10.10.1.20.



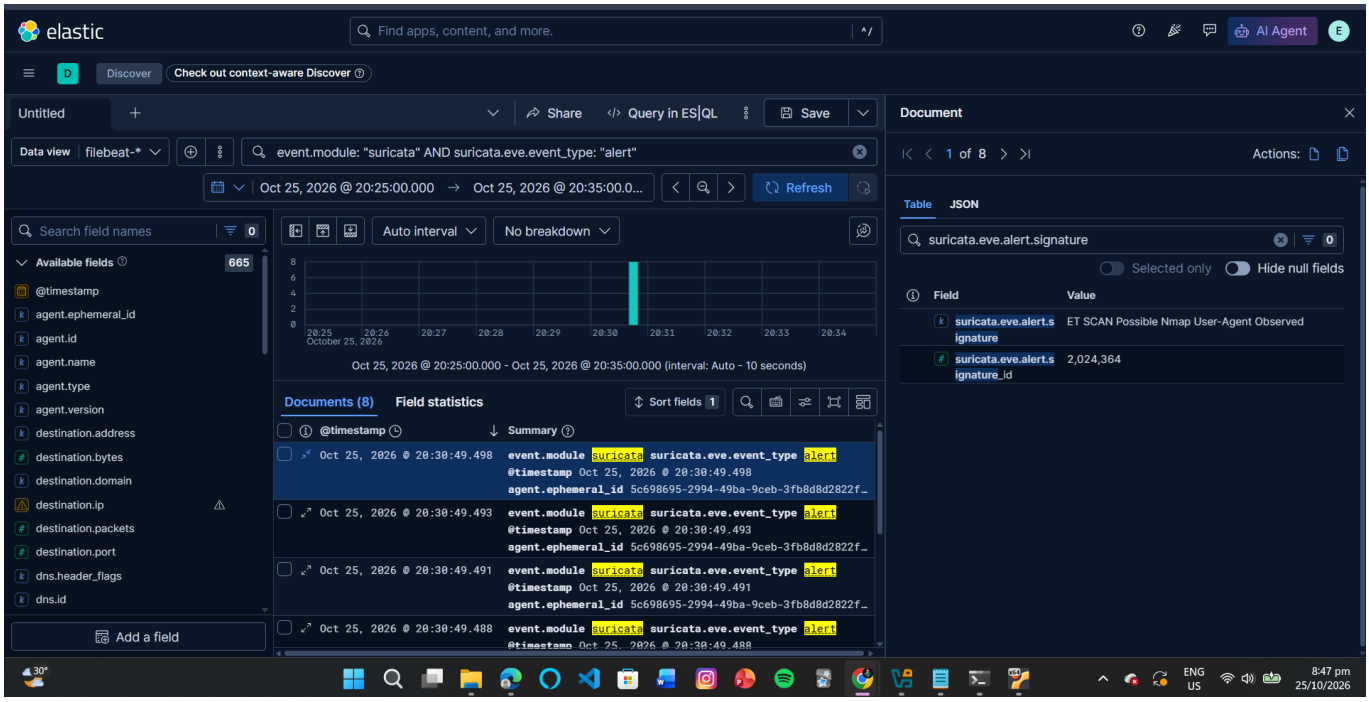
Kibana Discover: Suricata network telemetry involving the scanned host.

STEP 11B | Suricata scan alert and signature

I searched Suricata alert events around the scan time and identified an alert with signature ET SCAN Possible Nmap User-Agent Observed (signature ID 2024364).



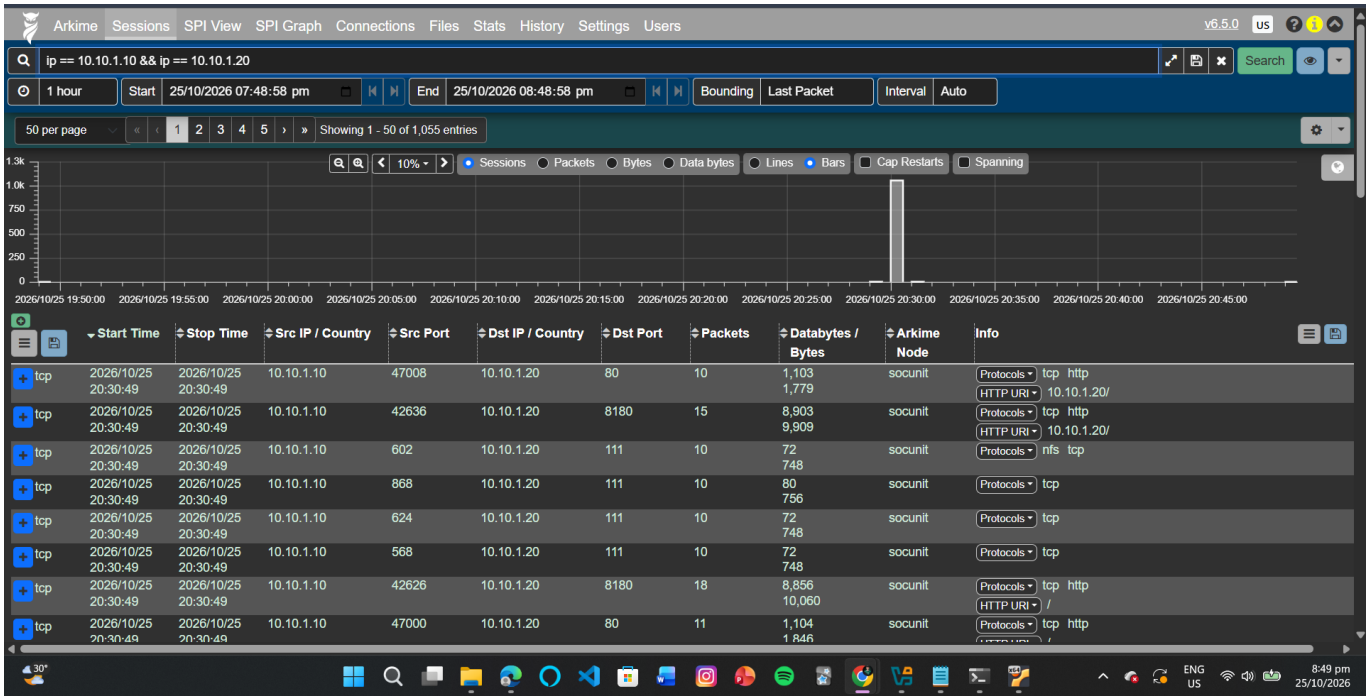
Kibana: eight Suricata alert events in the selected interval.



Expanded document showing the exact Nmap-related alert signature.

STEP 11C | Arkime captured scan sessions

I searched Arkime for sessions involving 10.10.1.10 and 10.10.1.20. The results showed 1,055 matching sessions, including TCP/HTTP traffic to ports 80 and 8180.



Arkime Sessions showing both laboratory IP addresses and matching traffic.

WRITTEN ANSWERS

1. What command did I use to set the static IP on the second interface?

I used **sudo ip addr add 10.10.1.10/24 dev eth1** to assign the internal network address. I also saved the static interface settings in /etc/network/interfaces for persistence.

2. What does PROMISC mean?

PROMISC means promiscuous mode. It permits a network interface to receive frames addressed to other devices when the virtual network delivers those frames. I set SIEMv2 Adapter 2 to Allow All.

3. What are three open ports shown in the Nmap scan?

I identified **21/tcp (FTP - vsftpd 2.3.4)**, **22/tcp (SSH - OpenSSH 4.7p1)**, and **80/tcp (HTTP - Apache httpd 2.2.8)**.

4. What alert signature appeared for my Nmap scan?

I observed the Suricata alert **ET SCAN Possible Nmap User-Agent Observed**, with signature ID **2024364**, in Kibana Discover.

Conclusion

I confirmed network communication between Kali and Metasploitable, identified open services through Nmap scanning, and correlated the test activity with Suricata alerts in Kibana and network sessions in Arkime.